UdbhavaX'26

- *Project Title: Farmer Crop Insurance Claim Tracking System – Bihar*
- *Domain: AG-01 (Agri Tech)*
- *Team Name: Code4Crops*
- *Team Members: D.Meghana, B.Spandana, T.Akhileshwari, S.Deepthi*

- *Department: CSE*
- *Event: Udbhava X'26, Coding Club SR University*

Problem Scenario

The Real-World Situation:

Agriculture is the backbone of rural livelihoods in states like Bihar. Thousands of farmers depend on seasonal crops for their income and survival. Natural disasters such as floods, droughts, and cyclones frequently damage crops, making crop insurance a critical support system for financial protection.

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The Current Failure:

When farmers apply for crop insurance claims, the process is largely manual and inefficient—relying on physical forms, office visits, and offline verification. There is no centralized digital system to manage or track claims. Farmers must repeatedly visit government offices to check their application status, leading to delays, confusion, and frustration. Missing documents, poor communication, and lack of transparency often result in delayed or rejected claims, increasing financial stress on farmers.

Current Pain Points (Why it Breaks)

- **No Centralized "Source of Truth":** Insurance claims are handled through multiple offline channels such as paper forms, local offices, and verbal communication. This makes it difficult to track the exact status of a claim or access complete information in one place.

- **The “Incomplete & Missing Data” Crisis:**

Farmers often submit applications with missing or incorrect documents due to lack of guidance. This leads to repeated re-submissions, delays, and increased workload for officers.

- **No Historical Data for Decision-Making:**

The absence of a digital record system prevents authorities from analyzing past claims, disaster patterns, and processing delays, making it difficult to improve the system over time.

Objectives & Success Criteria

Primary Objective:

To replace manual, paperwork-heavy insurance processes with a centralized digital platform that streamlines claim submission, improves transparency, and ensures efficient communication between farmers and authorities.

Success Criteria:

Successfully capture structured claim data (Farmer Details, Crop Type, Disaster Type, Location).

Enable farmers to upload and manage required documents digitally.

Provide real-time claim status visibility (Submitted → Under Review → Approved/Rejected).

Send automated notifications and reminders to farmers.

Allow officers to verify and approve claims efficiently.

Maintain complete claim history for transparency and tracking.

Enable authorities to monitor and manage claims through a dashboard view.

Proposed Solution Strategy

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Input Standardization
Structured digital forms for claim submission to prevent errors

Role-Based Interfaces
Separate dashboards for Farmers and Officers

Automated Claim Tracking
Real-time updates to keep farmers informed

Document Management System
Secure document upload, storage, and verification

Notification & Reminder System
Automated alerts via SMS/App to inform farmers

Centralized Data Management
Unified database for farmer records, claims, and history

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Key Solution Strategies:

1.Input Standardization:

Enforcing structured digital forms for claim submission to avoid errors, missing data, and confusion.

2.Role-Based Interfaces:

Separate dashboards for Farmers (claim submission & tracking) and Officers (verification & approval).

3.Automated Claim Tracking:

A system that provides real-time updates on claim progress without requiring physical visits.

System Architecture

Architectural Overview (3-Tier):

- **Frontend Layer (Web/App):**

Separate, user-friendly dashboards for **Farmers** and **Officers/Administrators**.

. Farmers: Claim submission, document upload, status tracking k

. Officers: Claim verification, approval, and monitoring

. Responsive design for mobile and low-bandwidth usage.

- **Backend Logic Layer:**

.Handles core system operations including:

. Data validation and form processing

. User authentication (Farmer / Officer login)

- Claim processing and status management
- Document verification workflow
- Notification handling (SMS/App alerts)

- **Data Persistence Layer**:

Stores structured and secure data for efficient access and tracking:

- Farmer records (personal & land details)
- Insurance claims and status history
- Uploaded documents and verification data
- Disaster type and location-based records
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Data Flow / Workflow

Submission: Farmer enters claim details (Crop Type,

Location,Disaster Type) and submits the application.

Validation Logic: System checks submitted data anddocuments.

Decision: If incomplete → Stop & Alert. If valid → Proceed

Visualization: Admin dashboard updates immediately. Claims are displayed with status (Pending/Approved/Rejected) and priority (e.g., disaster severity)

Implemented Features (Reference Prototype):

Role-based Login (Student/Admin).

Structured Complaint Reporting with Dropdowns.

Smart Duplicate Detection System.

Admin Dashboard with Real-time Updates.

Visual Priority Highlighting (Red for High Priority).

Issue Tracking with Status Visibility.

LINK REQUIREMENT:

GitHub Repo: https://github.com/DONTHIMEGHANA/sru-coding-hackathon/upload/main

Live Demo: https://farm-flow-bihar.lovable.app/farmer/dashboard

Impact & Benefits

For Students:

- Transparency and visibility of complaint status.

Eliminates repeated reporting of the same issue.

For Administrators:

- Reduces workload by preventing duplicate complaints.
- Enables quick identification of high-priority issues.

For the University:

- Provides structured data for better decision-making.
- Improves overall system efficiency and management.

Limitations

Prototype Data Persistence:

For this prototype, the system uses basic/local storage for data handling. Production deployment would require a secure cloud database.

Authentication:

Implements a basic login system; does not yet support advanced authentication like SSO or secure identity integration.

***Offline Capability*:**

System requires internet connectivity; full offline data submission and later synchronization is not yet implemented.

Future Scope & Roadmap

Phase 2: Intelligence & Mobile

AI-Based Claim Verification: Implement AI/ML models to automatically verify documents and detect fraud or inconsistencies in claims.

Mobile App Development: Build a Flutter/React Native app for farmers with features like photo upload (crop damage proof) and GPS-based location tagging.

Phase 3: Smart Integration

Government System Integration: Connect with official agriculture and insurance databases for faster validation and approval.

Satellite & Weather Data Integration: Use real-time weather and satellite data to automatically validate disaster events (floods, droughts).